

Smart Community Governance using Real-Time Priority Intelligence

1st Mrs.M.Shereesha

Computer Science and Engineering
B.V.Raju Institute of Technology
Narsapur,Medak,India
shereesha.m@bvrit.ac.in

2nd Bagam Charan

Computer Science and Engineering
B.V.Raju Institute of Technology
Narsapur,Medak,India
24211a0543@bvrit.ac.in

3th Baari Sneha

Computer Science and Engineering
B.V.Raju Institute of Technology
Narsapur,Medak,India
24211a0539@bvrit.ac.in

4th Alluri Lakshmi Preethi

Computer Science and Engineering
B.V.Raju Institute of Technology
Narsapur,Medak,India
24211a0515@bvrit.ac.in

5th A.Harshavardhan

Computer Science and Engineering
B.V.Raju Institute of Technology
Narsapur,Medak,India
24211a0516@bvrit.ac.in

6rd L.Pallavi

Computer Science and Engineering
B.V.Raju Institute of Technology
Narsapur,Medak,India
pallavi.lanke@gmail.com

Abstract—Smart Community Governance through Real Time Priority Intelligence project is a web-based solution used in improving the management of communities. The system enhances efficiency and organization in managing various communities simultaneously. The application has three roles assigned to its users namely Super Admin, Admin and Resident. The system uses JSON Web Token to ensure user authentication during login process. The technique ensures that the users are authenticated without any session information stored on the server. The system has also incorporated a way of handling community complains based on an algorithm that considers frequency and waiting time of each complain. This means that the urgent issues get taken care of faster. Smart Community Governance using Real-Time Priority Intelligence also has features, such as sending out emergency notifications managing events and keeping track of funds in a transparent way. These features help people in the community talk to each other better be more accountable and manage things overall. Smart Community Governance using Real-Time Priority Intelligence is really, about making residential communities better places to live.

Index Terms—Residential Community, Complaint Management System, Web Platform, User Management, Emergency Notifications.

I. INTRODUCTION

People who live in communities often have a hard time dealing with complaints and things like community events. They also have to collect money and respond to emergencies. Usually people do these things by hand or in a way. This means it takes a time to get things done and people do not always know what is going on. It is also hard for residents and the people in charge to talk to each other. A digital platform could make things easier for communities. The Smart Community Governance System is a website that residents can use. Residents can use the Smart Community Governance System to file complaints. They can also use the Smart Community Governance System to ask for updates on their complaints. The Smart Community Governance System

will let residents know when something changes in their community. This way residents can use the Smart Community Governance System to stay informed, about what's happening in their community.

II. LITERATURE SURVEY

In a system where there are various types of roles like Super Admin, Admin, and Resident, Role-Based Access Control (RBAC) is essential for proper functioning. RBAC, which was suggested by Ravi Sandhu and standardized by NIST, provides an effective way of assigning permissions on the basis of roles and not people [1], [2], [11]. The application of this concept to our system will allow us to manage access to various modules.

Authentication and safe communication are vital in any modern web application. The JSON Web Token (JWT) specification proposed by Michael Jones allows for safe transfer of user information from the client to the server [3], [12]. This technique also makes it possible to manage sessions in such a way that only authorized persons get access to protected resources.

There has been a lot of research in the domain of grievance redressal systems, particularly in hostels. The grievance redressal systems that are discussed in [4], [5], [6], and [7] make it clear that digitization of complaints is necessary in order to handle them properly. This type of a system allows for submission of complaints through the web and tracking of progress, thus providing the basis for our module. Moreover, further improvements to be made in complaint management systems would be the inclusion of prioritization techniques. Queue models for service issue prioritization [9] facilitate complaint processing based on their urgency and significance, which is a concept supported by our approach to calculating complaint priority using time and vote.

Complaint systems available via the web, such as those described in [18] and [19], also focus on user-friendly

interfaces and real-time tracking capabilities. Decision-making rules based on expert systems that effectively manage complaints are introduced in [20].

A crucial issue related to our complaint management system would be the development of its secure web architecture. The findings from research conducted by K. Lee provide insight into creating efficient and secure web applications [10].

Research studies like [17] prove the effectiveness of online grievance systems in facilitating communication between users and administration. Such platforms minimize manual efforts and ensure transparency, which is one of the aims of our Smart Community Governance System.

III. EXISTING WORK

There are several different kinds of management systems available that help address grievances and related issues, particularly within hostels and other institutional setups. The traditional grievance redressal systems were manual, which led to delays, ineffectiveness, and lack of transparency in addressing complaints. Web-based complaint management systems were introduced to address these problems. Some examples of web-based grievance management systems are presented in [5], [6], and [7], where users could make a digital complaint, and administrators could then track and resolve those. The research on hostel grievance management systems, such as [4], [8], and [9], was concerned with developing a centralized complaint portal, through which students could file complaints regarding various facilities. These systems offered complaint filing and tracking functionalities, but they did not provide for role-based access control and authentication. Role-Based Access Control (RBAC) is a system model proposed by Ravi Sandhu in [1] and [11]. The main objective behind introducing RBAC is that permissions can be assigned to a user according to the role played by the user in the process. Earlier grievance management systems did not incorporate RBAC properly. The use of techniques like JSON Web Token (JWT), suggested by Michael Jones [3], [12], increased the level of secured communication in web applications. Nevertheless, most of these applications use basic authentication techniques and therefore prone to possible security threats. Recently published works like [10] emphasize the importance of secure architecture of web applications. Furthermore, some publications, such as [17], prove that web-based grievance management systems increase effectiveness and quality of communication. Some advanced systems also focus on prioritization techniques used for handling grievances. Techniques like queuing model discussed in [9] play an important role in managing grievances according to their priority levels. Unfortunately, such functionality is not available in all basic systems. Overall, existing systems serve as a basis for managing grievances but have drawbacks including lack of real-time communication functionality, poor role-based management capabilities, insufficient prioritization

capabilities, and lack of up-to-date UI/UX design.

IV. LIMITATIONS IN EXISTING WORK

In spite of the availability of several online systems designed for management of complaints, there are numerous shortcomings in current methods. First, most of the systems have not adopted the use of real-time communication, and thus the interactions between users and administrators are cumbersome. Secondly, most of these systems do not utilize Role-Based Access Control, leading to poor permission management and hence insecurity of the systems. Again, some of the authentication systems used in these systems are weak and insecure and thus may expose the systems to risks. Thirdly, the problem of absence of a proper approval workflow for administrators and residents poses a serious threat in terms of controlling the system access. Fourthly, there is no proper system for managing complaint prioritization, and all complaints are managed on the first come first serve basis. Also, community-based data separation is not adopted by most of the current systems, exposing them to problems of data mixing. Some of the systems are old-fashioned and poorly designed in terms of the user interface. Again, there are no good notification systems. In addition, transparency is not very high since it is not easy for the users to monitor the status of their complaints. Lastly, some of the applications have scalability problems that hinder them from supporting more than one user effectively.

V. PROPOSED SYSTEM

The suggested Smart Community Governance System is a web-based application that enables efficient management of activities within the community, following the concept of role-based administration. The three users suggested for this system are: Super Admin, Admin, and Resident. Various capabilities provided by the system include user authentication, approvals, complaints processing, events management, announcement management, and fund management. Priority-based complaint processing has been incorporated in this design to ensure effective resolution of complaints, according to their level of urgency. Data isolation has also been achieved by ensuring community-specific

VI. LIST OF MODULES

- **The Smart Community Governance System (SCGS):** The Smart Community Governance System is built with a design. This helps it grow easily and work smoothly. The system has parts, each doing a specific job. This makes it easy for residents, administrators and system services to work together.
- **The User Authentication Module:** That is because the user authentication module handles user registration and login processes. This guarantees that the correct users will be allowed into the system. Another feature provided by

the user authentication module is code verification for the community.

- **The Community Management Module:**Community Management. The individual who establishes a community is automatically assigned the title of Super Administrator in that community. Additional administrators may join the community only by knowing the community code.
- **The Admin and Approval Management Module**is responsible for managing approvals in the system. Super Admins can. Reject requests from admins. Admins can. Reject requests from residents. This ensures that the system is secure and fair.
- **Complaints can be filed by residents:**Complaints from citizens on various aspects of the community may be filed. Complaints are then documented by the Complaint Management module, along with the priority levels assigned to them. Administrators can view, manage, and update complaints.
- **The Announcement Management Module:**This allows Super Admins and Admins to create announcements. The residents will be able to view these announcements. This helps keep everyone updated on the events happening in the community.
- **The Event Management Module:**helps organize and manage community events. Admins can configure event settings. Enable residents to sign up for the events. Residents can view events. Sign up if permitted.
- **The Fund Management Module:**tracks the community budget. Only the administrators have access to this module. The module provides information on, community budget.
- **When dealing with the Emergency Alert Module:**Provides real-time alerts. It helps the community act promptly.
- **The Database Management Module:**stores all system data. This comprises user information, complaints, events, and financial records. It ensures data is secure and easily accessible.
- **Modules in the SCGS:** These all modules work together. Each module serves a particular purpose. This ensures scalability, maintenance, and efficiency in the system. The SCGS helps communities manage activities. It provides an transparent way to handle community operations. The system is designed to be user-friendly. It helps residents, administrators and system services work together smoothly.

VII. SYSTEM ARCHITECTURE

- **Smart Community Governance System:**Smart Community Governance System is an architecture-driven solution which incorporates several layers that contribute to effective management and security of the software. The architecture of Smart Community Governance System contains four layers, namely the User Layer, Application Layer, Core Processing Layer, and Service Layer. There are particular tasks assigned to each layer, thus allowing users (Super Admin, Admin, and Resident) to get an excellent experience using the system.
- **The User Layer:** The User Layer comprises all participants who interact with the system. They include Super Admins, Admins, and Residents that have certain responsibilities in the system. Super Admin is responsible for managing the system and its communities. At the same time, Admins work within particular communities and manage their activities, for instance, handle complaints, organize events, and make decisions regarding various requests submitted by residents. Residents use the system to submit their complaints and attend particular events. The Application Layer deals with user requests and offers basic services such as authentication, complaints handling, event handling, announcement handling, and fund handling.
- **The Application Layer:** makes sure that any request by the user is processed correctly and the desired operations are carried out. The Application Layer creates an interface between the users and system logic so that the system can become interactive for the user.
- **The Core Processing Layer:** The Core Processing Layer is referred to as the SCGS Engine. The Core Processing Layer consists of various modules such as role manager, approval engine, notification engine, and priority engine. These modules ensure that there is secure access, the process of approvals is done, notifications are sent to the concerned authorities, and complaints are prioritized based on their urgency. Finally, the Service Layer ensures the proper handling of data storage and retrieval from databases related to users, complaints, events, and finance management.
- **Mathematical Equation**
The proposed system uses a math model to prioritize complaints in a way. It does not just handle them in the order they come in. The system looks at how important, urgent and impactful on the community each complaint's. This way it handles the critical ones first. To calculate the priority of each complaint the system uses things like how people vote for it how long its been waiting and

how severe it is. It adds these things together in a way to get a single score.

The score is based on things, like: How often people vote. How long the complaint has been waiting. The complaints severity level also plays a role.

Priority = Vote Frequency + Waiting Time

VIII. METHODOLOGY

- 1) **User Registration and Mapping to Community:** This module takes care of user registration and mapping of users to their respective communities using community codes.
 - User Role Selection (Admin/Resident)
 - Community Code Creation/Entry
 - User Mapping to Their Respective Communities
- 2) **Security and Authentication:** Security in this module comes from verification of user credentials along with role-based access control.
 - Email and Password Validation
 - Secured Login
 - Role-Based Access Control (RBAC)
- 3) **Role-Based Access Control (RBAC):** In this module, users' access to the application is managed using the approval process.
 - Super Admin Approvals of Admin Requests
 - Approval of Resident Request by Admin or Super Admin
 - Users with access approvals have permission to access the application.
- 4) **Complaint Handling:** Complaint handling is done using features available in this module.
 - Filing of Complaints by Residents
 - Admin Response to the Complaint
 - Update of Complaints Tracking and Updates
- 5) **Event and Announcement Management:** This is where communication takes place within the community.
 - Events created by admins
 - Events created by admins
 - Notices for information dissemination
- 6) **Core Processing (SCGS Engine):** This module takes care of the logic processing in the system.
 - Role management for access permissions
 - Approval engine for request processing
 - Notification process in real time
- 7) **Priority Handling Mechanism:** This module facilitates prioritized processing of complaints through a priority system.
 - Priority = Time + (Votes × 2)
 - Higher priority complaints processed first
 - Makes complaint management efficient and fast
- 8) **Data Management:** This module holds all the data needed for the application.
 - Storage of user, complaint, and event data

- Community-wise data segregation

- 9) **Output and System Response:** This is where the outputs from the modules are communicated to users.

- Complaints status shown on the interface
- Notifications and alerts
- Dashboard display of outputs

IX. RESULT

- 1) **User Authentication and Approval:** The user authentication system is designed to authenticate the user and ensure that the user is an approved one before accessing the portal. The system is developed such that only approved users can gain access.

- Authentication and authorization mechanism
- Email and password authentication

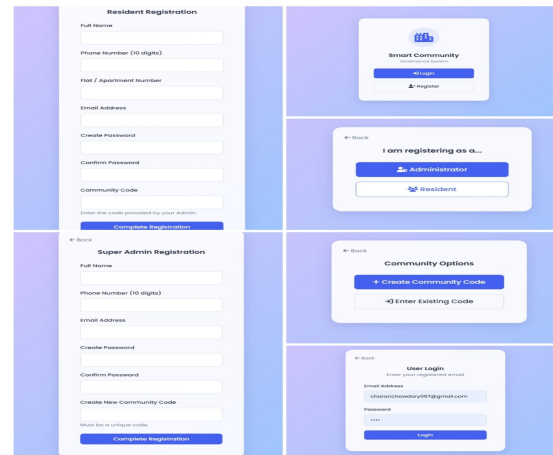


Fig. 1. Login and Registration Page

- 2) **Complaint Management:** The complaint management system will enable the residents to lodge their complaints and the administration to respond to these complaints via a chatting method.
 - Complaint submission via chatting
 - Administration response through chatting
 - Status of complaint (Pending/Resolved)
- 3) **Complaint Handling Prioritization:** Priority will be assigned to complaints in order to make the processing more efficient. The priority value will be determined based on time and number of votes.
 - Priority Value = Time + (Votes × 2)
 - Prioritize urgent complaints first
 - More efficient
- 4) **Alert and Notification:** Notifications enable real-time updates concerning what is happening in the system.
 - Complaint and Approval notifications
 - Real-time updates

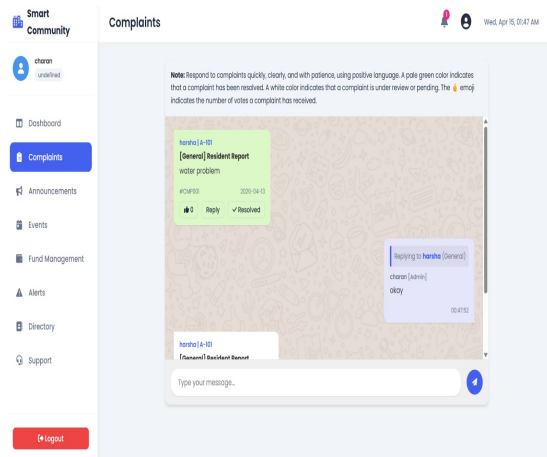


Fig. 2. Complaint Block

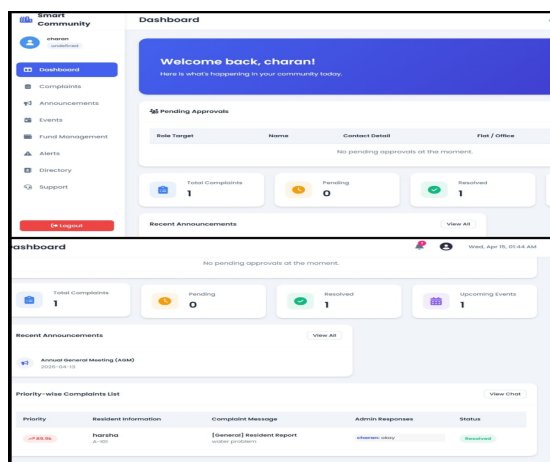


Fig. 3. Priority-wise Complaints

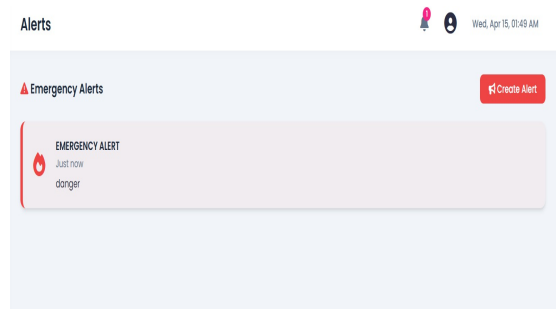


Fig. 4. Emergency Alert

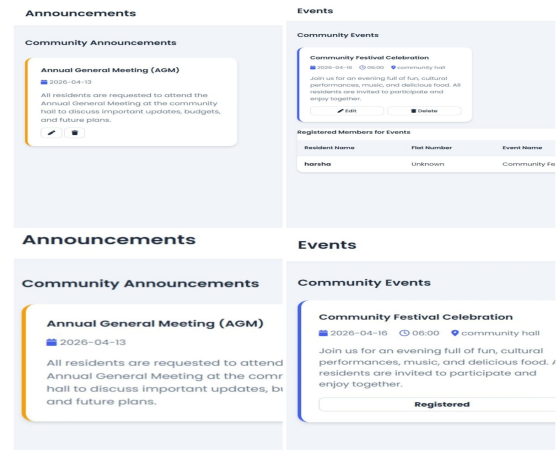


Fig. 5. Announcements and Events

5) **Announcement and Event Management:** This module facilitates announcements and events management by the admin. It enables residents to see announcements and join events without hassles.

- Admin creates announcements/events
- Residents can view and join events
- Improves communication

6) **Fund Management Module:** This module makes financial information visible to all users for tracking community funds.

- Displays income and expenses
- Increases transparency
- Includes QR payments

7) **Role-Based Access Control (RBAC):** This system uses RBAC for secure and smooth operation. Only authorized access to each feature is provided to respective users.

- Different roles to users

- Access control is done
- Increases system security

• **Community-Based Data Management:** This system handles multiple communities through effective data management for the purpose of maintaining privacy.

- Data separated community wise
- Data is safely stored
- Data never overlaps

• **Overall Performance of System:** The overall system runs effectively as a result of proper working of modules together

- Fast reaction time
- User friendly system

REFERENCES

- [1] R. Sandhu, D. Ferraiolo, and R. Kuhn, "Role-Based Access Control Models," IEEE Computer, vol. 29, no. 2, pp. 38–47, Feb. 1996. Osteoporos Int 2022; 33: 169–176
- [2] D. Ferraiolo, R. Sandhu, S. Gavrila, D. Kuhn, and R. Chandramouli, "Proposed NIST Standard for Role-Based Access Control," ACM Transactions on Information and System Security, vol. 4, no. 3, pp. 224–274, Aug. 2001.

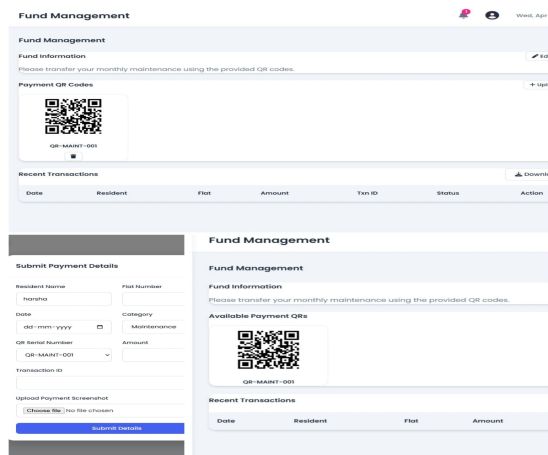


Fig. 6. Fund Management

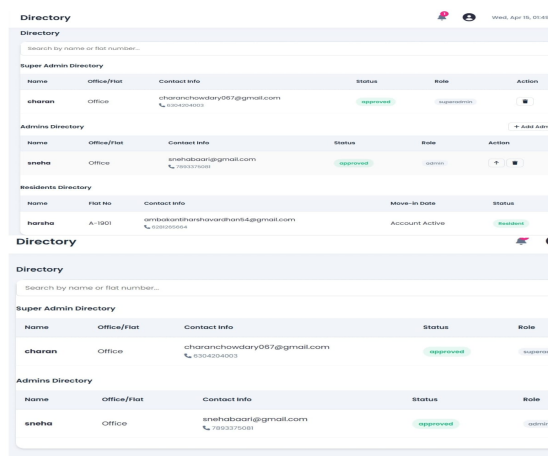


Fig. 7. Directory

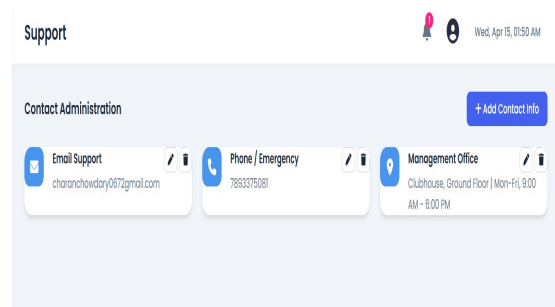


Fig. 8. Support Page

- [3] M. Jones, J. Bradley, and N. Sakimura, "JSON Web Token (JWT)," IETF RFC 7519, May 2015. 4.Kasar M. K. et al., "Hostel Grievance System," International Scientific Journal of Engineering and Management, 2023.
- [4] N. Kumar, S. Singh, and S. Gupta, "Design and Implementation of Grievance System for Hostel," in Proc. IEEE ICCCS, 2020.
- [5] M. K. Gupta, "Development of a Web-based Grievance Redressal System for Hostels," in Proc. IEEE COMITCon, 2019.
- [6] S. K. Sharma and A. Singh, "An Online Grievance Redressal System for Hostels," in Proc. IEEE SPIN, 2019.
- [7] N. A. Shaikh, N. S. Shaikh, and S. S. Shinde, "Hostel Complaint Management System," in Proc. IEEE I2CT, 2018.
- [8] A. D. Chougale and A. D. Chougale, "Hostel Management System with Grievance Redressal Facility," in Proc. IEEE ICCIC, 2014.
- [9] P. R. Bhandari et al., "Hostel Complaint Management System," in Proc. IEEE ICCPCT, 2016. 11.J. Wang et al., "Service Issue Prioritization Using Queue-Based Models," in Proc. IEEE International Conference on Services Computing, 2017.
- [10] .K. Lee and H. Kim, "Secure Web Application Architecture Design," Journal of Web Engineering, vol. 20, no. 3, pp. 455–472, 2021.
- [11] Sandhu R. et al., "Role-Based Access Control Models," IEEE Computer, 1996.
- [12] Jones M. et al., "JSON Web Token (JWT) Standard," IETF RFC 7519, 2015.
- [13] .Sharma S. K. and Singh A., "Online Grievance Redressal System," IEEE SPIN, 2019.
- [14] Shaikh N. A. et al., "Complaint Management System Architecture," IEEE I2CT, 2018.
- [15] Kumar N. et al., "Web Portal Complaint Framework for Hostels," IEEE ICCCS, 2020.
- [16] Kasar M. K. et al., "Digital Grievance Redressal Model for Hostel Management," International Scientific Journal of Engineering and Management, 2023.
- [17] U. N. Tripathi, A. K. Srivastava, and B. P. Singh, "Effectiveness of Online Grievance Redressal and Management System: A Case Study of IGNOU Learners," Indian Journal of Educational Technology, vol. 3, no. 2, pp. 92–105, 2021.
- [18] Sneha Alve et al., "Web Application for Complaint Tracking and Resolving," International Journal of Innovative Science, Engineering and Technology, 2017.
- [19] Osman Nasr and Enayat Alkhider, "Online Complaint Management System," International Journal of Innovative Science, Engineering and Technology, 2015.
- [20] Y. Helmy, "A Decision Support Model to Improve Complaint Handling Using Rule-Based Inference Engine," Journal of Theoretical and Applied Information Technology, 2024.